Parallel Architectures and Programming Models, 2026W

Assignment 1: OpenMP Tasking, Roofline Model

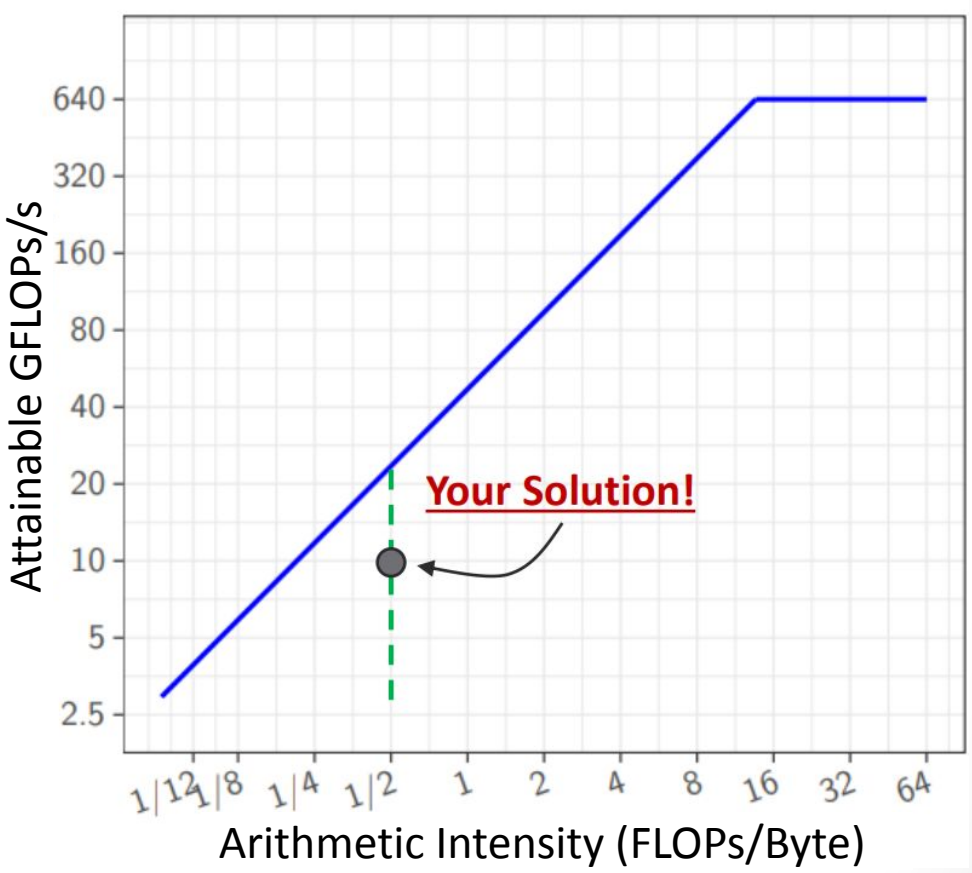
University of Vienna

GPSA :: Roofline Model

Assignment Objectives:

1) Plot the **roofline model** for a single **ALMA node** in the **Charts** section of the **submission platform**.
2) Fill in the value boxes for **Q17–Q19** on **page 3** of the **Quiz**, using your best-performing GPSA parallel kernel with **32 threads** to process the input strings in **X2.txt** and **Y2.txt**.
3) Add the Arithmetic Intensity and Performance of the previously selected kernel to the **Roofline Model** in the **Charts** section.
4) Fill in the value boxes for **Q20–Q21** on **page 3 of the Quiz**, using the results from the plots obtained in **steps 1 and 3**.

Example answer for points 1) and 3):



About ALMA Specifications…

One node of ALMA consists of 2 Intel Xeon E5-2650 CPUs :

- **each with 2.0 GHz clock rate.**

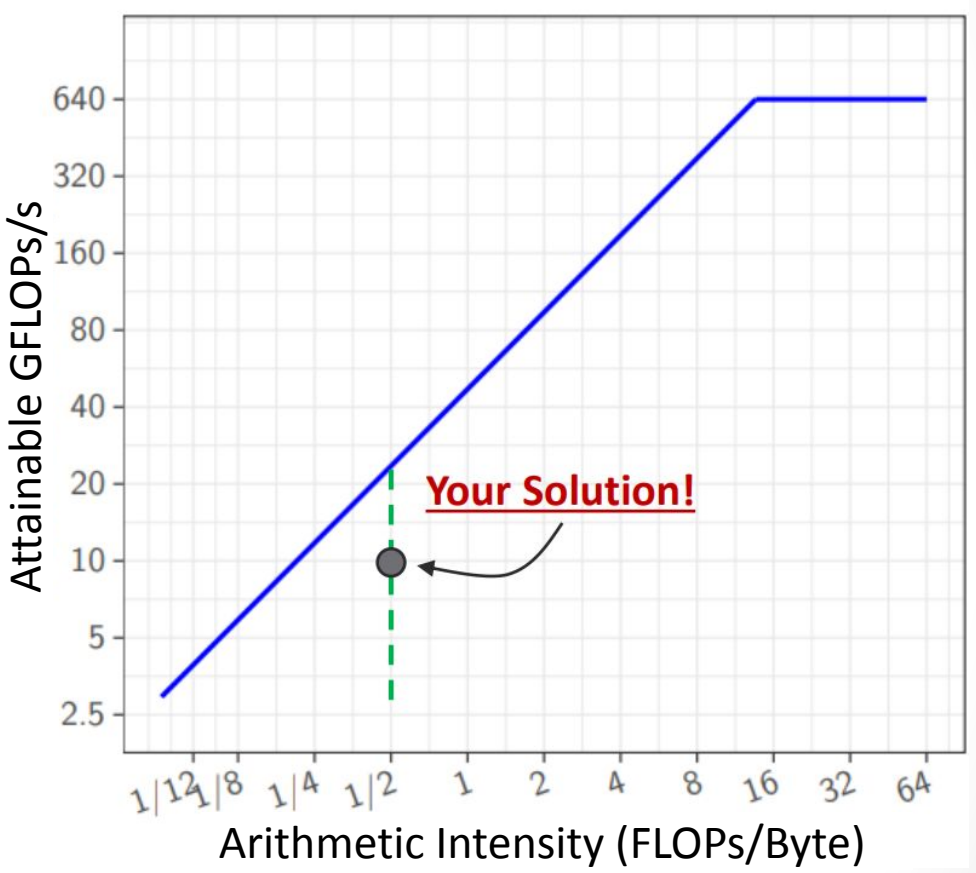
Each CPU has 8 physical cores:

- **each core can execute up to 2 AVX instructions per cycle:**
  - `_mm_256_add_ps`: **8 single-precision FLOPs (ADD)**
  - `_mm_256_mul_ps`: **8 single-precision FLOPs (MUL)**

The memory bandwidth of each ALMA node is 57 GB/s.

Finally, assume that `sizeof(float)` is equal to 4 bytes.

Example answer for points 1) and 3):



About GPSA Arithmetic Intensity…

To compute the Arithmetic Intensity (AI) of your parallel solution (32 threads), you should only consider the code inside the innermost-loop of GPSA:

- **for simplicity's sake ignore initialization loops.**
- **if you modified the innermost-loop, calculate the Arithmetic Intensity (AI) of the code on left instead.**

Consider only floating-point operations (FLOPs):

- **ignore operations with `X` and `Y` → those are `chars`.**
- **ignore operations with loop counters → those are `ints`.**

Assume `std::max({match,del,insert})` needs 2 FLOPs.

GPSA kernel:

```
for (unsigned int i = 1; i < rows; i++) {
  for (unsigned int j = 1; j < cols; j++) {
    float match = S[i-1][j-1] +
                  (X[i-1] == Y[j-1] ?
                   match : mismatch);
    float del = S[i-1][j] + gap_penalty;
    float insert = S[i][j-1] + gap_penalty;
    S[i][j] = std::max({match, del, insert});

    visited++;
  }
}
```

Additional Information

Reminder: To earn a positive grade on Assignment 1, make sure you complete *both* tasks.

All information required to plot the roofline model is provided in "About ALMA specifications…",

Or can be obtained by analyzing and/or running your parallel solution on ALMA:

- **e.g., #FLOPs executed, execution time, …**

You should NOT use profilers to obtain information:

- **this includes PAPI, PERF, LIKWID, etc...**

The deadline for this part of the assignment is the same as for the first part of the project!